BTS-IV-04.19-0314 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

A

B.Tech. Degree IV Semester Examination April 2019

ME 1402 METROLOGY AND INSTRUMENTATION

(2012 Scheme)

Time: 3 Hours Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Describe the methods for estimating accuracy and precision.
(b) Briefly explain gauge maker's tolerance.
(c) Explain the working principle of an auto collimator.
(d) Write a note on optimeters.
(e) Distinguish between systematic error and random error.
(f) Briefly explain classification of measuring instruments.
(g) Give a brief account on total radiation pyrometer.
(h) Briefly explain the working of scintillation counter.

PART B

(4 × 15 = 60)

II. (a) Describe the necessity and importance of metrology. Also explain system of limits and fits. (7)
(b) Briefly explain line standard and end standard. (8)

OR

III. (a) Explain hole basis and shaft basis system of representation. (7)
(b) Write notes on: (8)
(i) Classification of gauges
(ii) Gauge meterials

IV. Explain the uses of sine bar with a neat sketch. (15)

OR

V. (a) Explain the construction and working of tool maker's microscope. (7)
(b) State the adverse effects of poor surface finish. Give any four symbols for direction of lay. (8)

VI. (a) Describe the working of mercury in glass thermometer as a first order instrument. (7)
(b) With diagram explain functional elements of an instrument. (8)

OR

VII. (a) Explain the working of seismic instrument as a second order instrument. (7)
(b) Discuss the generalized mathematical model of measuring systems of second order instruments. (8)

VIII. With neat sketch explain the working of: (15)
(i) ORSAT's apparatus
(ii) Hydraulic load cell

OR

IX. Briefly explain: (15)
(i) Bonded strain gauges
(ii) Strain rosettes
(iii) Acoustical measurements

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